

Notes:

- Cheating or Trying to Cheat may disqualify you from the course.
- ChatGPT or any AI tool is not allowed in this exam. Use only HI (Human Intelligence)!

Use the following instruction to login to 'Linux server' using Windows PowerShell

`ssh <academic-user-name>@172.31.80.155`

Q1) (15 Points)

Write a deterministic C program that simulates a Ping-Pong game using threads, mutex, and condition variables. The game should run for 4 rounds.

In each game round, the main thread should create two player threads: one that prints "Ping" and another that prints "Pong". The threads must alternate correctly, with "Ping" always preceding "Pong" in each round. Use mutex locks and condition variables to ensure proper synchronization between the threads.

After each game round, the main thread should print "Round X completed" (where X is the round number). After all 4 rounds are completed, the main thread should print "All games finished!"

Solution Instructions:

1. Write the solution on a C file (game.c)
2. Compile and run each program from your terminal.
3. Wait for grading.